

5.3.8 External clock timing characteristics

5.3.8.1 High-speed external user clock generated from an external source

In bypass mode, the HSE oscillator is switched off and the input pin is a standard GPIO.

The external clock signal has to respect the I/O characteristics in I/O port characteristics. However, the recommended clock input waveform is shown in the figure below.

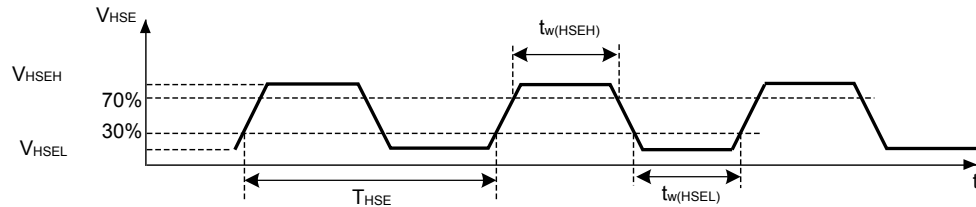
Table 33. High-speed external user clock characteristics

Specified by design and not tested in production.

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
f_{HSE_ext}	User external clock source frequency	Digital mode (HSEYBYP = 1, HSEEXT = 1)	-	-	50	MHz
		Analog mode (HSEYBYP = 1, HSEEXT = 0)	4	-	50	
V_{HSEH}	OSC_IN input pin high-level voltage	Digital mode (HSEYBYP = 1, HSEEXT = 1)	$0.7 \times V_{DD}$	-	V_{DD}	V
V_{HSEL}	OSC_IN input pin low-level voltage	Digital mode (HSEYBYP = 1, HSEEXT = 1)	V_{SS}	-	$0.3 \times V_{DD}$	
$t_{w(HSEH)}^{(1)}$ $t_{w(HSEL)}^{(1)}$	OSC_IN high or low time	Digital mode (HSEYBYP = 1, HSEEXT = 1)	7	-	-	ns
$DuCy_{HSE}$	OSC_IN duty cycle	Digital mode (HSEYBYP = 1, HSEEXT = 1)	45	-	55	%
$V_{HSE_ext_PP}^{(2)}$	OSC_IN peak-to-peak amplitude	Analog mode (HSEYBYP = 1, HSEEXT = 0)	0.2	-	$2/3 V_{DD}$	V
V_{HSE_ext}	OSC_IN input range		0	-	V_{DD}	
$t_{r(HSE)}, t_{f(HSE)}$	OSC_IN rise and fall time		$0.05 / f_{ext_ext}$	-	$0.3 / f_{ext_ext}$	ns

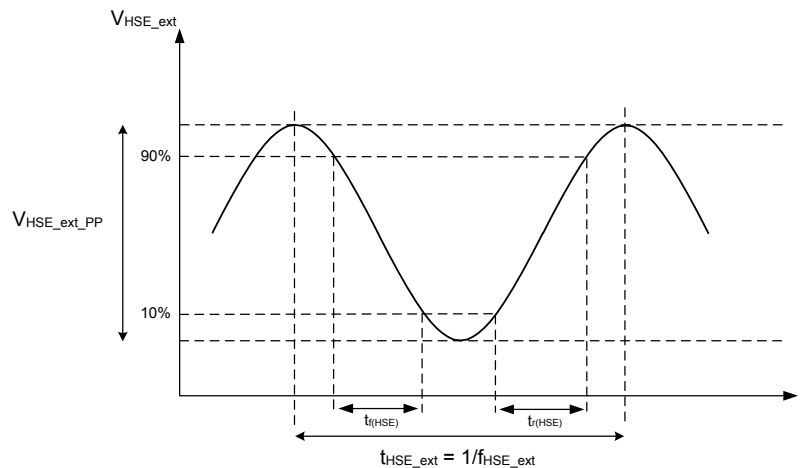
1. There is no specified rise and fall time for a digital input signal, but the V_{HSEH} and V_{HSEL} conditions must be fulfilled.
2. The DC component of the signal must ensure that the signal peaks are located between V_{DD} and V_{SS} .

Figure 14. AC timing diagram for high-speed external clock source (digital mode)



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Figure 15. AC timing diagram for high-speed external clock source (analog mode)



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